

Planted, But Never Prosperous: India's Struggle with Low Agricultural Yield

Whenever I go on road trips in India, I often pass through vast farms with long rows of crops, usually tended by traditional farmers. I not only admire the crop's beauty and outstanding produce but also marvel at the journey of a lonely seed to an integral part of my plate. However, it breaks my heart to see the other face of Indian agriculture – a farmer bending on his back and using a sickle, his crops barely fitting a basket, and his children missing school to give a helping hand.

Eventually, I realized that India's most pressing development issue is its low agricultural productivity and wide yield gap. Most farmers in India grow for sustenance, eating the harvested crop and trading the little that's left in local markets, never reaching a buyer willing to pay its true worth.

The Economic Paradox

According to the Periodic Labor Force Survey Annual Report, around 43-46% of Indians are involved in the primary sector for their livelihood, and these activities contribute to just 16% to 18% of the nation's GDP! In contrast, only about 1.2% of the US Labor force works directly in farming, yet American agriculture remains amongst the world's most productive because of scientific farming.

The Roots of the Problem

Today, nearly 85% of Indian farmers are small or marginal, meaning that the land they own is just about 5 American football fields. One major reason is land fragmentation, which makes it nearly impossible to practice industrialized farming.

India's lack of efficient machinery and its anachronistic techniques limit biomass yield. Sickles and wooden plows are common tools that take an eternity to complete a task. Disguised employment pushes farmers to have even less incentive to buy machinery, as they have "free" workers when more families do a task than are needed.

From Failed Crops to The Poverty Cycle

Depressed harvest volumes lead farmers to not have enough money to fulfil their needs, leading to seed debt, malnutrition and the illiteracy of their children; the main reasons to make someone get stuck in the poverty cycle. Some farmers move to cities to get low-paying jobs, and this separates families and crowds the cities. This development issue also spikes cost-push food inflation and global imports, and a wage-price spiral forms.

The Path Forward

Petty crimes and celebrity gossip dominate major headlines, but this space is truly needed to address the unnoticed battle of Indian farmers. This issue also includes rising food prices, rural poverty, migration, and India's future and is not only about farmers.

American youth can launch educational campaigns in schools and neighborhoods. Tech-savvy students can develop open-source apps using satellite data to predict weather or detect crop diseases and design affordable Arduino-based soil sensors for grassroots organizations to replicate locally.

American philanthropies and lending platforms like Kiva can pool diaspora capital with Indian microcredit NGOs to ease farmers' debt and help them join Farmer Producer Organizations (FPO) that sell directly to fair-trade buyers, eliminating middlemen profits.

For reliable water, which Indian farmers gamble on, India must shift to smart-and-precise irrigation. Tech companies and universities can share automated ag-tech knowledge to help create low-cost, localized solutions for Indian farms.

Solving this alone is impossible. Let's unite; together, we can raise awareness, encourage people, and bring others in until the burden is shared by many shoulders. If we were facing this same problem, what would we hope the literate world would do for us? We need to act.

If not us, who?

If not now, when?

Sources

1. Government of India, Ministry of Statistics and Program Implementation. *Periodic Labor Force Survey Annual Report 2025*. Used for labor-force and workforce context.
2. Government of India, Ministry of Agriculture and Farmers Welfare. *Agricultural Statistics at a Glance 2024–25*. Used for agriculture’s contribution to India’s economy and related agricultural statistics.
3. Government of India, Department of Agriculture and Farmers Welfare. *Agriculture Census 2015–16*. Used for data on small and marginal farmers and landholding patterns.
4. USDA Economic Research Service. *Agriculture and Its Related Industries Provide 10.4 Percent of U.S. Employment*. Used for comparison with U.S. direct on-farm employment.
5. USDA Economic Research Service. *Ag and Food Statistics: Charting the Essentials — Farming and Farm Income*. Used for information on U.S. productivity growth, mechanization, and farm-sector transformation.
6. World Bank Blogs. *Unlocking Agricultural Transformations for Small and Marginal Farmers in Uttar Pradesh, India*. Used for discussion on irrigation efficiency, mechanization, and productivity improvements.